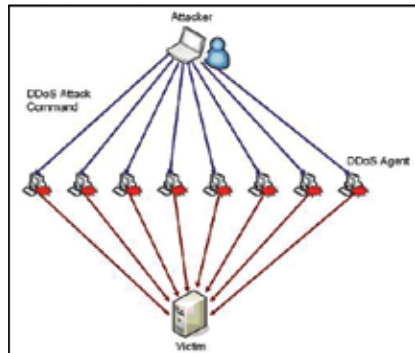


that during peak traffic time, Facebook processes more than 500K (that's 5 lakhs) requests per second. Sending it a few more thousand requests per second would not make a lot of difference. For such cases, there is another, more dangerous method: DDoS.

DDoS or Distributed DoS is the power of DoS multiplied n number of times. The number n is the number of



DDoS - Brute Force attack, the networking style

computers involved in DoS. Confused? Well, DDoS distributes the overall DoS load onto multiple computers which parallelizes the effort and multiplies it. We can think of it like this: If 5K requests to Facebook do not bring the servers down, let us get 1000 computers and make each of them fire 5K requests to Facebook servers per second. In such a case, you would be sending $5000 \times 1000 = 5,000,000$ requests per second. Now, that might just bring down Facebook if its servers cannot take that many requests! The toughest part of DDoS is simply getting that many computers.

In many cases DDoS is performed by an army of Zombie computers which are already under control. These computers may be anything from a smartphone to a server as long as they are connected to the internet and can be commanded to send huge number of requests per second. To create such an army, yet another step comes in - to create a virus which can be used for creating backdoors for commanding, controlling and/or scheduling the DDoS attack. However, since cyber warfare is usually pre-planned by a group, the availability of computers may not be a problem.

One of the biggest strengths of DDoS is that not only can it flood the servers, it can also bring networking equipment to a halt. Typically, when DoS is detected (well, there are systems for that too), networking equipments are configured to block any requests coming from the attacking IP address. If there are a few computers located on the same network which are attacking the target, a blocking rule can be set on a complete IP address range. As the number of computers increase along with the diversity of IP addresses which are being used by the sources (computers) of attack, the rules get more complex. Let us consider the previous example of attacking Facebook servers: